

JOSEPH KEHINDE, BOLUJO

Hardware Engineer | Embedded Systems Developer

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PROFESSIONAL SUMMARY

Computer Engineering student specialized in Embedded Systems and UAV Architecture. Designing flight controllers from the ground up, with proficiency in **PlatformIO**, **STM32Cube Software Suite**, and committed to building world-class hardware solutions on African soil, bridging the gap between low-level firmware and complex hardware integration.

TECHNICAL SKILLS

- **Microcontrollers (MCUs):** ESP32 (ESP-IDF & Arduino), STM32, Arduino Uno/Mega/Nano.
- **Development Environments:** VS Code + PlatformIO, ESP-IDF, Arduino IDE, Vivado (FPGA/SoC).
- **Engineering Tools:** PCB Design (KiCad), Proteus.
- **Software Stack:** C/C++, Python.
- Interest in researching and reading on trends about SBCs, MCUs, and drone technology.

CORE TECHNICAL PROJECTS

Smart Parking Lot System (ESP32) | *Completed*

- Designed and simulated an automated parking ecosystem using ESP32 to manage occupancy data via IoT protocols.
- Optimized system power consumption and sensor accuracy through rigorous simulation and testing.

UAV Flight Controller (Final Year Project) | *In Progress*

- Architecting a custom flight controller for UAV systems, supporting both brushed and brushless motor configurations.
- Implementing PID control loops and sensor fusion algorithms to ensure flight stability and real-time response.
- Developing firmware using C within the STM32Cube Suite and C++ within the PlatformIO environment for optimized hardware utilization.
- Additional functionalities being implemented are telemetry viewing via a web-based dashboard and remote control via Wi-Fi and cellular networks, control using a remote control transmitter, mapping, and autonomous control (among others).

- Evaluating performance with a custom-made drone and comparing it with industry-standard flight controllers

Custom Quadcopter (Final Year Project) | *In Progress*

- Designing and assembling a custom drone for evaluating my custom flight controller.
- Designing and implementing a custom Power Distribution Board (PDB) to power various aspects of the drone efficiently.

SoC Security Research: Logic Locking | *In Progress*

- Conducting a technical review of Logic Locking techniques to safeguard Intellectual Property (IP) in Systems-on-Chip (SoC).
- Simulating hardware security protocols in Vivado to evaluate the robustness of locking mechanisms against piracy.

EXPERIENCE

Redeemer's University Technology Space | Vice President *September 2025 – Present*

Constantlink Technologies Limited | Hardware Engineering Intern *April 2025 – July 2025*

- Diagnosed and repaired complex PCB-level issues for smartphones (Samsung, Infinix, Oppo).
- Engineered solar power solutions, including load calculation and panel/inverter installation for residential clients.
- Maintained industrial-grade hardware, including inverters, UPS systems, and printers.

MTN Nigeria | Network Intern *August 2025 – September 2025*

- Assisted in monitoring network infrastructure and troubleshooting telecommunications hardware.
- Carried out site audits as a component of maintenance and troubleshooting activities.

EDUCATION

Redeemer's University Bachelor of Engineering (B.Eng.), Computer Engineering *January 2021 – October 2026 (Expected)*

- **Key Coursework:** Computer Architecture, Microprocessor Systems, Digital Logic Design, Embedded Systems, Control Systems, Cryptography Principles and Applications.